

Exact Spectral Inheritance in Prime-Power Resolution Towers: A Splitting Theorem for Affine-Cocycle Transfer Operators, with Application to the Collatz Carry Sum

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Abstract

For a prime power $q = p^e$, the accelerated Collatz carry sum generates a multiplicative walk on $K_e = \langle 2, 3 \rangle \leq (\mathbb{Z}/q\mathbb{Z})^\times$ and an associated family of phase-twisted transfer operators $\mathcal{L}_{e,t,x}$, whose dominant twisted eigenvalue controls the rate of equidistribution of carry-sum residues at fixed p (prior work: *Fixed-Prime Equidistribution of Accelerated Collatz Carry Sums*). This note settles the natural follow-up question — how this spectral picture behaves as the resolution e is refined — by proving an exact structural theorem for the whole tower $(\mathcal{L}_e)_{e \geq 1}$. We show that the natural pullback $J_e : \mathcal{H}_e \rightarrow \mathcal{H}_{e+1}$ along the reduction $\pi_e : K_{e+1} \rightarrow K_e$ not only intertwines the operators ($\mathcal{L}_{e+1}J_e = J_e\mathcal{L}_e$, under a single generic hypothesis on the multiplicative order of 2) but that its image is a *reducing*, not merely invariant, subspace: the orthogonal complement Z_{e+1} (functions of zero sum on every fiber of π_e) is *also* exactly invariant, giving an orthogonal direct-sum splitting $\mathcal{L}_{e+1} \cong \mathcal{L}_e \oplus \mathcal{N}_{e+1}$. This is strictly stronger than the block-upper-triangular structure a bare invariant subspace would give, and it has an immediate consequence that a weaker structural statement could not deliver: the spectral-gap quantity governing equidistribution, $g_0(p, e)$, is *unconditionally non-increasing* in e — resolving, in the monotone-and-convergent direction, an open question raised by preliminary numerical data that had suggested (incorrectly, as it turns out) an ongoing decay. We identify exactly which arithmetic hypothesis the theorem needs (generic multiplicative-order lifting of the base 2), verify it fails at a concrete, named prime (a Wieferich prime base 2), and give a precise account of why a *different* arithmetic anomaly at a different prime (Wieferich base 3) perturbs the spectral data without threatening the theorem’s hypothesis at all. The theorem and its proof are stated for a general affine-cocycle operator family, not specifically for the Collatz instance; what remains open is a sharper, correctly-isolated question about whether finer, stratum-local spectral dominance always holds, which is not needed for the monotonicity result itself.

Scope of claims. *Proved here, generically (Theorems 3, 6, 9):* the intertwining identity; the exact orthogonal splitting; monotonicity and convergence of $g_0(p, e)$. *Proved here, number-theoretically (Prop. 13):* the equivalence of the Fermat-quotient and order-stall forms of the Wieferich condition, and its exact role in the hypothesis. *Verified computationally (exact/symbolic, small cases):* the splitting identity and the eigenvalue-union formula, for several primes and a non-Collatz control multiplier. *Explicitly open:* whether a non-primitive stratum's new-sector operator can ever exceed its own local inherited eigenvalue (Open Problem 12); whether the limit of $g_0(p, e)$ is strictly positive. *Not claimed:* anything about Collatz cycle closure, divergence, or the Collatz conjecture — this note is purely about the statistical/spectral structure of the carry-sum residue system.

1 Introduction

Fixed-Prime Equidistribution of Accelerated Collatz Carry Sums [1] proves that, for the accelerated Collatz carry sum $C_L(D) = \sum_{j < L} 3^{L-1-j} 2^{s_j}$ over valuation words D , the residue $C_L(D) \bmod p$ equidistributes as $L \rightarrow \infty$, for every fixed prime $p \nmid 6$, at an explicit geometric rate governed by a transfer-operator spectral gap $g_0(p)$. The companion instrument *amgap* [2] extends the construction to prime powers $q = p^e$ and reports two data points suggesting the gap might shrink with e : $g_0(11, 1) = 0.3340$, $g_0(11, 2) = 0.3225$, explicitly flagging the question of whether $g_0(p, e) \rightarrow 0$ as $e \rightarrow \infty$ as open.

A preliminary numerical extension (unpublished working notes) computed $g_0(11, 3) \approx 0.305$, apparently continuing the downward trend, and an initial structural analysis of the prime-power tower proposed a recursion $B_p(e) = \max\{P_p(e), B_p(e-1)\}$ (with $P_p(e)$ the dominant primitive-stratum eigenvalue and $B_p(e)$ the overall gap-defining quantity) based on an embedding of lower-level spectra into higher levels. On audit, that recursion was found to rest on an unproved assumption — that an embedded (“inherited”) eigenvalue remains globally dominant within its own stratum — and the monotonicity conclusion was downgraded to conditional.

This note resolves the question. The key missing structural fact is that the natural reduction map's pullback is not merely order-preserving (an invariant subspace) but *splits* the operator as an honest orthogonal direct sum. This upgrades the available inequality to an exact formula, from which unconditional monotonicity follows by a short, elementary argument (Theorem 9). We also locate, precisely, the one arithmetic hypothesis the whole tower construction needs (Section 6), and the one concrete prime where it is known to fail.

2 The operator family

Definition 1 (Affine-cocycle transfer operator). *Fix a prime p and coprime integers $a, b \geq 2$ with $p \nmid ab$. For $e \geq 1$, let $q = p^e$ and $K_e = \langle a, b \rangle \leq (\mathbb{Z}/q\mathbb{Z})^\times$, a cyclic group (every subgroup of $(\mathbb{Z}/p^e\mathbb{Z})^\times$ is cyclic for odd p). Let $r_e = \text{ord}_q(b)$. For $t \in \mathbb{Z}/q\mathbb{Z}$ and x in the open unit disk, define $\mathcal{L}_{e,t,x} : \mathcal{H}_e \rightarrow \mathcal{H}_e$ on $\mathcal{H}_e := \mathbb{C}^{K_e}$ by*

$$(\mathcal{L}_{e,t,x} f)(u') = \sum_{u \in K_e} e_q(tu) \frac{x^{d_0(u,u')}}{1 - x^{r_e}} f(u),$$

where $e_q(y) = \exp(2\pi i y/q)$ and $d_0(u, u') \in [1, r_e]$ is the least d with $u' = u \cdot b^d \cdot a^{-1} \bmod q$. The Collatz instance is $(a, b) = (3, 2)$, with $K_e = \langle 2, 3 \rangle$ the reachable states of the multiplicative walk $u_{j+1} = u_j b^{D_j} a^{-1}$ underlying $C_L(D) \bmod q$ [1], and the relevant evaluation point is the saddle $x_* = 1 - 1/\log_2 b$ (for $(a, b) = (3, 2)$: $x_* = 1 - 1/\log_2 3$), at which $\rho(\mathcal{L}_{e,0,x_*}) = \log_2 b - 1$ exactly, independent of p, e [1].

We fix the normalized inner product $\langle f, g \rangle_e := \frac{1}{|K_e|} \sum_{u \in K_e} f(u) \overline{g(u)}$ throughout (uniform probability measure on K_e), the unique normalization making the resolution maps below isometric.

Definition 2 (Resolution map). $\pi_e : K_{e+1} \twoheadrightarrow K_e$ is reduction mod p^e , restricted to K_{e+1} ; this is a surjective group homomorphism with constant fiber size $m := |K_{e+1}|/|K_e|$. $J_e : \mathcal{H}_e \rightarrow \mathcal{H}_{e+1}$, $J_e f := f \circ \pi_e$ (pullback); $S_e : \mathcal{H}_{e+1} \rightarrow \mathcal{H}_e$, $(S_e f)(\bar{u}) := \sum_{u \in \pi_e^{-1}(\bar{u})} f(u)$ (fiber-sum).

A direct computation shows J_e is injective and, with the normalization above, an exact isometry; $S_e \circ J_e = m \cdot \text{id}$.

3 Intertwining and the splitting theorem

Theorem 3 (Pullback intertwining). Suppose (H1): $r_{e+1} = p \cdot r_e$. Then for every $t \in \mathbb{Z}/p^e \mathbb{Z}$,

$$\mathcal{L}_{e+1,pt,x} J_e = J_e \mathcal{L}_{e,t,x}.$$

Proof. Fix source $u \in K_{e+1}$, target class $\bar{u}' \in K_e$. The valid increments d taking u to some $u' \in \pi_e^{-1}(\bar{u}')$ are exactly $\{d_0^{(0)} + kr_e : k = 0, \dots, p-1\}$, where $d_0^{(0)} \in [1, r_e]$ solves the coarser congruence at level e (using (H1) to know this set has exactly $p = m$ elements inside $[1, r_{e+1}]$); the phase $e_{p^{e+1}}(ptu) = e_{p^e}(t\bar{u})$ is constant on the fiber. Summing:

$$\sum_{u' \in \pi_e^{-1}(\bar{u}')} (M_{pt}(x))_{u',u} = e_{p^e}(t\bar{u}) \sum_{k=0}^{p-1} \frac{x^{d_0^{(0)} + kr_e}}{1 - x^{r_{e+1}}} = e_{p^e}(t\bar{u}) \frac{x^{d_0^{(0)}} (1 - x^{pr_e})}{(1 - x^{r_e})(1 - x^{r_{e+1}})} = e_{p^e}(t\bar{u}) \frac{x^{d_0^{(0)}}}{1 - x^{r_e}},$$

using $r_{e+1} = pr_e$ in the last step — exactly $(M_t(x))_{\bar{u}', \bar{u}}$. Summing against any f constant on fibers gives the claim. $\square$

Remark 4. The proof never used $\gcd(t, p) = 1$; it lifts any residue t at the lower level by one factor of p , not only “primitive” ones.

Lemma 5 (Pushforward intertwining). Under (H1), $S_e \circ \mathcal{L}_{e+1,pt,x} = \mathcal{L}_{e,t,x} \circ S_e$.

Proof. Identical computation, read for the fiber-sum of the *output* index against a fixed, arbitrary source: $(S_e \mathcal{L}_{e+1} f)(\bar{u}') = \sum_u f(u) \sum_{u' \in \pi_e^{-1}(\bar{u}')} (M_{pt})_{u',u} = \sum_u f(u) (M_t)_{\bar{u}', \pi_e(u)} = \sum_{\bar{u}} (M_t)_{\bar{u}', \bar{u}} \sum_{u \in \pi_e^{-1}(\bar{u})} f(u) = (\mathcal{L}_e S_e f)(\bar{u}')$. $\square$

Theorem 6 (Splitting). Under (H1), $Z_{e+1} := \ker(S_e)$ satisfies:

1. $Z_{e+1} = J_e(\mathcal{H}_e)^\perp$ exactly;
2. Z_{e+1} is invariant under $\mathcal{L}_{e+1,pt,x}$;
3. $\mathcal{H}_{e+1} = J_e(\mathcal{H}_e) \oplus Z_{e+1}$ is an orthogonal direct sum of two $\mathcal{L}_{e+1,pt,x}$ -invariant subspaces, and in an orthonormal basis respecting this sum,

$$\mathcal{L}_{e+1,pt,x} \cong \mathcal{L}_{e,t,x} \oplus \mathcal{N}_{e+1,pt,x}, \quad \mathcal{N}_{e+1,pt,x} := \mathcal{L}_{e+1,pt,x}|_{Z_{e+1}}.$$

Proof. (1): $\langle h, J_e \bar{g} \rangle_{e+1} = 0$ for all $\bar{g}$ iff $\sum_{u \in \pi_e^{-1}(\bar{u})} h(u) = 0$ for every $\bar{u}$, i.e. $S_e h = 0$. (2): if $S_e f = 0$ then $S_e(\mathcal{L}_{e+1} f) = \mathcal{L}_e(S_e f) = 0$ by Lemma 5. (3) is immediate from (1)+(2): two orthogonal, each-invariant subspaces summing to the whole space give an exactly block-diagonal matrix in any adapted orthonormal basis. $\square$

This is strictly stronger than the invariant-subspace statement alone (Theorem 3 gives only block-upper-triangularity, for which the complementary block need not be invariant under the adjoint). Theorem 6 was verified independently by exact symbolic computation for $p = 5, 13$ (Section 7), including for *non-primitive* source residues t , confirming the theorem's generality beyond the primitive case.

Corollary 7. $\text{Spec}(\mathcal{L}_{e+1,pt,x}) = \text{Spec}(\mathcal{L}_{e,t,x}) \sqcup \text{Spec}(\mathcal{N}_{e+1,pt,x})$ (multiset union, full Jordan structure inherited), and

$$\rho(\mathcal{L}_{e+1,pt,x}) = \max(\rho(\mathcal{L}_{e,t,x}), \rho(\mathcal{N}_{e+1,pt,x})) \quad \text{exactly.}$$

Corollary 8. The full singular-value spectrum of $\mathcal{L}_{e+1,pt,x}$ is exactly the union of the singular-value spectra of $\mathcal{L}_{e,t,x}$ and $\mathcal{N}_{e+1,pt,x}$ (since the adjoint is block-diagonal in the same orthonormal basis).

4 Monotonicity

Write $B_p(e) := \max_{t \neq 0} \rho(\mathcal{L}_{e,t,x_*})$, $g_0(p, e) := 1 - B_p(e)/\lambda_0$ ($\lambda_0 = \rho(\mathcal{L}_{e,0,x_*})$, constant in e).

Theorem 9 (Monotonicity). Suppose (H1) holds at the step $e \rightarrow e+1$. Then $B_p(e+1) \geq B_p(e)$, hence $g_0(p, e+1) \leq g_0(p, e)$.

Proof. Let t_* achieve $B_p(e) = \rho(\mathcal{L}_{e,t_*,x_*})$ (exists, finite set; t_* need not be primitive). By Corollary 7 applied to the lift pt_* : $\rho(\mathcal{L}_{e+1,pt_*,x_*}) = \max(B_p(e), \rho(\mathcal{N}_{e+1,pt_*,x_*})) \geq B_p(e)$. Since $B_p(e+1)$ is a maximum over all nonzero residues including pt_* : $B_p(e+1) \geq \rho(\mathcal{L}_{e+1,pt_*,x_*}) \geq B_p(e)$. $\square$

Corollary 10. $g_0(p, \cdot)$ is a non-increasing, nonnegative sequence (nonnegativity: $\rho(\mathcal{L}_{e,t,x_*}) \leq \rho(\mathcal{L}_{e,0,x_*}) = \lambda_0$ for every t , since $|M_t| = M_0$ entrywise), hence convergent.

Remark 11. This does not determine the limit. Two logically separate questions remain open: whether only finitely many levels e set a new record for B_p (equivalently, whether the limit is reached at some finite level), and whether the limit is strictly less than λ_0 ("mixing survives") or equal to it (" $g_0 \rightarrow 0$ "). Theorem 9 answers neither; it only establishes that the sequence is monotone and therefore has some limit.

The proof of Theorem 9 isolates exactly what remains open:

Open Problem 12 (Stratum-local dominance). For a non-primitive t (writing $t = pt'$), is $\rho(\mathcal{N}_{e+1,pt',x_*}) \leq \rho(\mathcal{L}_{e,t',x_*})$, i.e. does the new sector ever exceed its own inherited value? This is not needed for Theorem 9 (which only requires one safe witness, the lift of the historical maximizer), but it is the natural finer question about the tower's internal structure, and is not resolved here.

5 Explicit model for the new sector

$Z_{e+1} = \ker(S_e)$ is concretely the space of functions on K_{e+1} with zero sum on every fiber of π_e . Since K_{e+1} is cyclic, fixing a generator g gives a canonical character basis χ_k ($k = 0, \dots, |K_{e+1}| - 1$, via discrete log); $J_e(\mathcal{H}_e)$ corresponds exactly to the χ_k trivial on $\ker \pi_e$, and Z_{e+1} to the remaining ones. When K_{e+1} is the full unit group ("index 1" primes), this coincides with the classical characters of exact conductor p^{e+1} ; when K_{e+1} is a proper subgroup, the identification with classical primitive Dirichlet characters is not automatic and is not claimed. The χ_k diagonalize $\mathcal{L}_{e+1,0,x}$ (a circulant in this basis) but not $\mathcal{L}_{e+1,t,x}$ for $t \neq 0$ in general; no closed form for $\mathcal{N}_{e+1,t,x}$'s eigenvalues at $t \neq 0$ was found. $\dim \mathcal{Q}_{e+1} = (p-1)|K_e|$ under generic growth — a constant fraction $(p-1)/p$ of the total dimension at every step, neither vanishing nor dominating.

6 Arithmetic anomalies

Proposition 13. *For $p \nmid a$: $a^{p-1} \equiv 1 \pmod{p^2}$ iff $\text{ord}_{p^2}(a) = \text{ord}_p(a)$.*

Proof. $\text{ord}_{p^2}(a) \in \{\text{ord}_p(a), p \cdot \text{ord}_p(a)\}$ always (standard order-lifting for prime powers); $\text{ord}_p(a) \mid p-1$ (Fermat) while $p \cdot \text{ord}_p(a) \geq p > p-1$ cannot divide $p-1$. So $a^{p-1} \equiv 1 \pmod{p^2} \iff \text{ord}_{p^2}(a) \mid p-1 \iff \text{ord}_{p^2}(a) = \text{ord}_p(a)$. $\square$

$p = 11$ (**Wieferich base 3**): $3^{10} \equiv 1 \pmod{121}$, verified directly; one of exactly two known such primes (11, 1006003). Direct order computation confirms $\text{ord}_{11}(2) = 10$ lifts generically ($\times 11$) at every level tested, so **(H1) holds throughout for $p = 11$** — Theorems 6–9 apply unconditionally, and the previously-observed three-level transient in $P_{11}(e)$ before freezing is consistent with, and now known to sit inside, a provably monotone sequence. The Wieferich-3 anomaly affects the *primitive*-stratum operator’s internal structure (there being no inherited part there to split against) and remains the reason for the specific numerical values of $P_{11}(e)$ — but it does not interact with the splitting mechanism at all.

$p = 1093$ (**Wieferich base 2**): $2^{1092} \equiv 1 \pmod{1093^2}$, verified directly; the smallest known base-2 Wieferich prime. Direct computation: $\text{ord}_{1093}(2) = 364$, $\text{ord}_{1093^2}(2) = 364$ (not 364×1093) — **(H1) fails exactly at $e = 1 \rightarrow 2$** . Notably, the group K_{e+1} itself still grows generically ($|K_2|/|K_1| = 1093$ exactly, since 3 is not simultaneously anomalous and compensates) — but this does *not* rescue Theorem 3’s proof, because the geometric-series telescoping is tied specifically to $\text{ord}(2)$ ’s own lifting, not to the group’s overall size. This is a precise illustration that (H1) is the correct, minimal, and non-redundant hypothesis.

7 Numerical verification

All verification in this note used exact symbolic (**sympy**) or high-precision numerical linear algebra on small cases, not broad numerical sweeps. Confirmed exactly: (i) $S_e M_{e+1} = M_e S_e$ and invariance of $\ker S_e$, for $p = 5, 13$ at $e = 1 \rightarrow 2$, both for primitive and non-primitive lifted residues; (ii) the exact block-diagonal structure (off-diagonal blocks $< 10^{-14}$ in a properly orthonormalized basis) and the resulting exact singular-value union (Corollary 8), matched to 10^{-8} across the six leading values; (iii) $\rho(\mathcal{L}_{\text{hi}}) = \max(\rho(\mathcal{L}_{\text{lo}}), \rho(\mathcal{N}))$ exactly, for both primitive and non-primitive source residues, $p = 5, 13$; (iv) the Wieferich computations of Section 6, by direct modular exponentiation; (v) genericity of the splitting mechanism for a non-Collatz control multiplier $(a, b) = (5, 2)$ at an arbitrary (non-saddle) evaluation point, confirming the theorem’s proof uses no Collatz-specific arithmetic. All scripts are included in **src**/.

8 Conclusion

The prime-power resolution tower of accelerated-Collatz-carry-sum transfer operators splits exactly, orthogonally, at every step satisfying one checkable arithmetic hypothesis. This upgrades a previously conditional monotonicity claim for the equidistribution spectral gap $g_0(p, e)$ to an unconditional theorem, and sharpens the remaining open question to a precise, stratum-local statement (Open Problem 12) that is no longer entangled with the coarser question the research program originally posed. The theorem and its proof are generic to affine-cocycle transfer operators of this shape; the Collatz carry sum is the motivating instance, not a hypothesis of any result proved here.

References

- [1] E. De Jesús, *Fixed-Prime Equidistribution of Accelerated Collatz Carry Sums: A Transfer-Operator Spectral Gap, with Two Open Uniformity Problems*, technical note, May 2026.

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